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Pediatric Eye-Related Emergency Department Visits and Pediatric Eye Provider Location and Density

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Objective: There is a shortage of pediatric eye specialists. Inaccessible eye care may lead parents to bring their children to the emergency department (ED) to address eye problems that could have been handled at a clinic visit. This study aimed to determine the association between childhood eye-related ED visits and the location and density of pediatric eye specialists in California.

Design: A population-based cross-sectional study.

Participants: All California ED visits between January 2012 and December 2021 for patients ≤18 years of age were identified using the California Office of Health Care Access and Information Database.

Methods: International Classification of Diseases diagnosis codes were used to identify eye-related ED visits. Public databases were used to identify pediatric ophthalmologist and optometrist addresses. Census data were used to determine the number of visits per 10 000 children in each zip code. Poisson regressions evaluated associations at the zip code level of eye-related ED visits and provider density with geographic sociodemographic factors.

Main Outcome Measures: Incidence of pediatric eye-related ED visits.

Results: Of the 117 363 721 ED visits in California between 2012 and 2021, 27 346 729 (23.3%) were for children, of which 391 985 (1.4%) were eye-related. Median age was 5.0 years (interquartile range: 2.0, 10.0), and 51.7% were male. The most common diagnoses were conjunctivitis (250 028; 63.8%), chalazion/blepharitis (56 389; 14.4%), and other disorders of the eye/adnexa (13 070; 3.3%). The mean number of visits per year per 10 000 children was 43 ± 12 (median 46, interquartile range: 43, 51). The estimated number of pediatric eye providers in California in 2023 was 142 (1 per 61 413 children). Zip codes with more pediatric eye providers had fewer eye-related ED visits: each additional provider per 10 000 children was associated with 2.1 fewer eye-related ED visits per 10 000 children (95% confidence interval −0.04 to −4.25; $P = 0.046$).

Conclusions: Most eye-related ED visits were for nonemergent eye conditions. There was a low eye provider-to-child ratio and an association between provider density and eye-related ED visit incidence. Expanding pediatric eye care could improve access and decrease ED utilization for nonemergent eye concerns.

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Supplemental material available at www.ophthalmologyscience.org.

Every year, millions of children receive care in emergency departments (EDs), and rates of ED utilization are increasing.[1,2] Many of these ED visits have low or minor acuity.[3] Receiving nonemergent care in the ED is inefficient as it contributes to longer wait times for patients with true emergencies and can lead to excessive health care spending.

One reason parents may choose to seek care in the ED when their child develops a nonemergent issue is limited access to outpatient pediatric care. Inefficient ED utilization may be particularly pronounced for those seeking subspecialty care in fields that have a low number of practicing providers. Theoretically, improving access to outpatient subspecialty providers could lead to a decreased reliance on ED visits.

Despite many medical trainees pursuing pediatric fellowships, there is an increasing shortage of access to pediatric subspecialty care throughout much of the United States.[4] Pediatric ophthalmology faces one of the most severe projected workforce shortages in the country, with just over 1000 providers in the United States.[5–7] Inadequate or inaccessible subspecialty eye care may lead parents to bring their children to the ED to address eye problems that could have been handled in an outpatient visit, if available. The goal of this study was to determine the association between childhood eye-related ED visits and the

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geographic location and density of pediatric eye specialists in California.

Methods

Dataset and Study Population

Data for this analysis were obtained from the California Department of Health Care Access and Information (https://hcai.ca.gov/data/request-data/) Emergency Department Dataset. The study included all records from patients ≤18 years of age who received care in a California ED between January 2012 and December 2021. This dataset comprises demographic, clinical, payer, and facility information from all hospitals licensed to provide emergency medical services in the state of California. Primary and secondary International Classification of Diseases diagnosis codes were used to identify eye-related ED visits (Table S1, available at www.ophthalmologyscience.org).

The Emergency Department Dataset was supplemented with data from several external datasets, merged based on the patient's residential zip code. These included Zip Code Tabulation Areas-level sociodemographic data from the 2017 US Census American Community Survey (ACS) via the tidycensus R package (https://www.census.gov/programs-surveys/acs/data/data-via-api.html, https://walker-data.com/tidycensus/), 2010 rural–urban commuting area codes (https://www.ers.usda.gov/data-products/rural-urban-commuting-area-codes/) (with scores ranging from 1 to 10, where higher scores indicate more rural locations), and yearly state-level demographic information from the ACS 1-Year Estimates Data Profiles (Table DP05) from 2012 to 2021. The 2017 ACS census data were used to determine the number of visits per 1000 children in each zip code.

Definitions and Conventions

Race/Ethnicity. The normalized race/ethnicity categories from the Emergency Department Dataset were used and largely retained as is. For the years 2019 to 2021, the Asian and Native Hawaiian/Pacific Islander categories were combined into a single category because they were recorded as 1 category before 2019. Payer: The expected source of payment was simplified into 3 categories: uninsured (self-pay), privately insured, and publicly insured. Language: The preferred language spoken at home was grouped into 5 categories: English, Spanish, other Indo-European languages, Asian and Pacific Island languages, and other languages (Table S2, available at www.ophthalmologyscience.org).

Pediatric Providers

On November 21, 2023, 4 publicly available databases were queried to estimate the number of pediatric ophthalmologists and pediatric optometrists in each zip code in California: the American Academy of Ophthalmology,[8] the American Association of Pediatric Ophthalmology and Strabismus,[9] the American Optometric Association,[10] and the American Academy of Optometry.[11] Provider addresses were cross-referenced with a Google search to confirm the physician's practice location.

Statistical Analysis

A univariate linear regression was performed at the zip code level to assess the relationship between pediatric ophthalmology ED visit prevalence and provider density for every 10 000 children by zip code. Five clear outlier zip codes were excluded before performing the regression. Poisson regressions were used to evaluate associations at the zip code level of eye-related ED visits and pediatric provider density with geographic sociodemographic factors. For the Poisson regression, the unit of analysis was the zip code. The outcome variable was the count of ED visits in each zip code, and the population of children in each zip code was used as an offset (log-transformed). The exposure variables were percentage of the population with a high school diploma, percentage of the population with a bachelor's degree, percentage of the population that is foreign-born, percentage of households where a non-English language is spoken at home, unemployment rate, median household income, percentage of households on food stamps or supplemental nutrition assistance program, percentage of families below the poverty line, rural–urban commuting area scores, and distance to closest pediatric provider. Distances were calculated at the zip code level, using the centroid from the shape file of each pediatric provider's zip code and the zip code of interest. For the zip code level regression and geographical analyses, we excluded out-of-state or invalid zip codes. All data analysis was conducted in R (version 4.3.3; The R Foundation for Statistical Computing, http://www.r-project.org/), and statistical significance was set at 0.05.

Geographic Analysis

The zctaCrosswalk (https://cran.r-project.org/web/packages/zctaCrosswalk/index.html) and tidycensus (https://walker-data.com/tidycensus/) R packages were used to obtain shapefiles for California Zip Code Tabulation Areas. The urbnmapr package was used to obtain shapefiles for each county (https://github.com/UrbanInstitute/urbnmapr). The geosphere R package (10.32614/CRAN.package.geosphere) was used to calculate distances between zip codes as defined by the centroid of each zip code shapefile's geometry. For graphing geographic analyses, we excluded zip codes that had total populations ≤100.

Results

Of the 117 363 721 ED visits in California between 2012 and 2021, 27 346 729 (23.3%) were for children ≤18 years, 391 985 (1.4%) were eye-related and included in the demographic and incidence analysis, and 384 795 were included in the zip code-level analyses after excluding out of state or invalid zip codes (Fig S1, available at www.ophthalmologyscience.org). For eye-related visits, the median age was 5.0 years (interquartile range: 2.0, 10.0), and 51.7% were male. The most common self-reported race/ethnicity was Hispanic (55.2%), primary language was English (82.7%), and insurance type was public (70.9%). Participant demographics are shown in Table 3. Conjunctivitis was the most common diagnosis, accounting for 250 028 (63.8%) of the eye-related ED visits, followed by chalazion/blepharitis (56 389 visits; 14.4%), and other disorders of the eye/adnexa (13 070 visits; 3.3%; Fig S2, available at www.ophthalmologyscience.org). Other less common conditions included ocular pain (10 081; 2.6%), subconjunctival hemorrhage (6153; 1.6%), other visual disturbances (4730; 1.2%), and orbital cellulitis (4621, 1.2%). On average, the incidence of pediatric eye-related ED visits was 43 per 10 000 children-years. The mean number of visits per year per 10 000 children was 43 ± 12 (median 46, interquartile range: 43, 51; Fig 3). The mean number of visits per year per 10 000 children was the highest for conjunctivitis

Table 3. Study Participant Demographics

Variable	Eye-Related Emergency Room Visits, n = 391 985	
	N	%
Sex		
Male	202 481	51.7
Female	189 494	48.3
Unknown	10	0.0
Age		
Median (IQR; yrs)	5.0 (2.0, 10.0)	
Language		
Asian and Pacific Island languages	2474	0.6
English	324 205	82.7
Other Indo-European languages	984	0.3
Other languages	1449	0.4
Spanish	62 873	16.0
Race/ethnicity		
American Indian/Alaska Native	1230	0.3
Asian/Pacific Islander	18 563	4.7
Black	49 000	12.5
Hispanic	216 346	55.2
Multiracial	936	0.2
Other	19 165	4.9
White	81 730	20.9
Unknown/invalid/blank	5009	1.3
Insurance type and status		
Public insurance	277 955	70.9
Private insurance	84 383	21.5
Uninsured	29 256	7.5
Other/unknown	391	0.1

IQR = interquartile range.

(27.6 visits per 10 000 children) and lowest for orbital cellulitis (0.5 visits per 10 000 children; Table 4).

The search of professional databases identified a total of 142 pediatric eye providers in the state of California in 2023. The US census estimated a population of 9 110 253 children ≤18 years of age in the state. Thus, the ratio of pediatric eye care providers to children in California was 1 per 61 413 children. Using the 2017 ACS estimates of demographic estimates per zip code, of the 1462 zip codes with a population of ≥100 children, 1371 (93.8%) zip codes in which 8 367 155 children reside (91.8% of the total pediatric population) had no pediatric eye care providers. Thirty-six of California's 58 counties (62.0%), in which 1 912 259 children reside (21.0% of the total pediatric population), have no pediatric providers. The highest density of providers by zip code was 94 720 (Berkeley, 7.1 per 100 children), and by county was San Francisco County (6.9 per 100 000 children; Fig 3).

The number of eye-related ED visits increased from 2012 to 2019 from 4.12 to 4.81 per 1000 children per year before declining sharply in 2020, likely due to the impacts of ED utilization from the coronavirus disease 2019 pandemic. Temporal trends were similar when stratified by sex, language spoken, and race/ethnicity (Fig S2). A strong seasonal trend was determined with a peak monthly incidence in March to April (Friedman rank test, QS test, Kruskall−Wallis, and Weber−Ollech test all $P < 0.001$).

Poisson regression revealed several zip code-level sociodemographic factors that were associated with an increased rate of eye-related ED visits for a given zip code, including distance to the nearest provider (Table 5). For provider density, the only zip code level predictive factors were percentage of residents with a bachelor's degree (rate ratio: 1.98 [1.17−3.33], $P = 0.01$) and median household income (incidence rate ratio: 0.80 [0.72−0.88], $P < 0.001$). There was a trend toward higher incidence of eye-related ED visits in zip codes that are farther away from the nearest pediatric provider (Fig 4A, B). We found that 50% of children live within 17.6 km of the nearest pediatric provider, 90% of children live within 102.0 km, and the furthest zip code is 263.1 km away from the nearest pediatric provider (Fig 4C). There was a statistically significant increase in the mean ED visit incidence between zip codes that were <30 km, between 30 and 50 km, and >50 km away from the nearest pediatric provider (all $P < 0.05$, Fig 4D). Univariate linear regression demonstrated that zip codes with more providers had fewer eye-related ED visits (Fig S5, available at www.ophthalmologyscience.org). Each additional pediatric provider per 10 000 children was associated with 40.3 (95% confidence interval: −80.3 to −0.4; $P = 0.048$) fewer ED visits per 10 000 children over the 10-year study period.

Discussion

We found that 1.4% of the 27 346 729 pediatric ED visits in California between 2012 and 2021 were for eye-related care, the majority of which were for nonemergent conditions. Zip codes with fewer pediatric eye providers had more eye-related ED visits.

The decision to seek care in the ED for nonemergent conditions is influenced by a myriad of factors. This study suggests that lack of access to an eye provider may contribute to higher rates of ED utilization for eye-related care among children. In our cohort, the most common conditions diagnosed in the ED were conjunctivitis and chalazion/blepharitis, neither of which threaten a patient's long-term vision nor are considered emergent. Of note, this study was based on diagnostic codes determined by ED providers, and it is possible that these may not be accurate. For instance, serious conditions such as uveitis or scleritis could be erroneously diagnosed as conjunctivitis. Providing care for these conditions by an eye provider not only has the potential to decrease cost but also can provide greater diagnostic accuracy and higher therapeutic quality of care.

We found that there was only 1 provider per 61 413 children in California, and each additional provider per 10 000 children was associated with 40.3 fewer eye-related ED visits per 10 000 children over the 10-year study period. Additionally, the geographic distribution of the practicing providers was skewed, with several areas of high provider density (often related to a university-based practice) and 1371 zip codes (91.8% of the total pediatric population) and 36 of 58 total counties (21.0% of the total pediatric

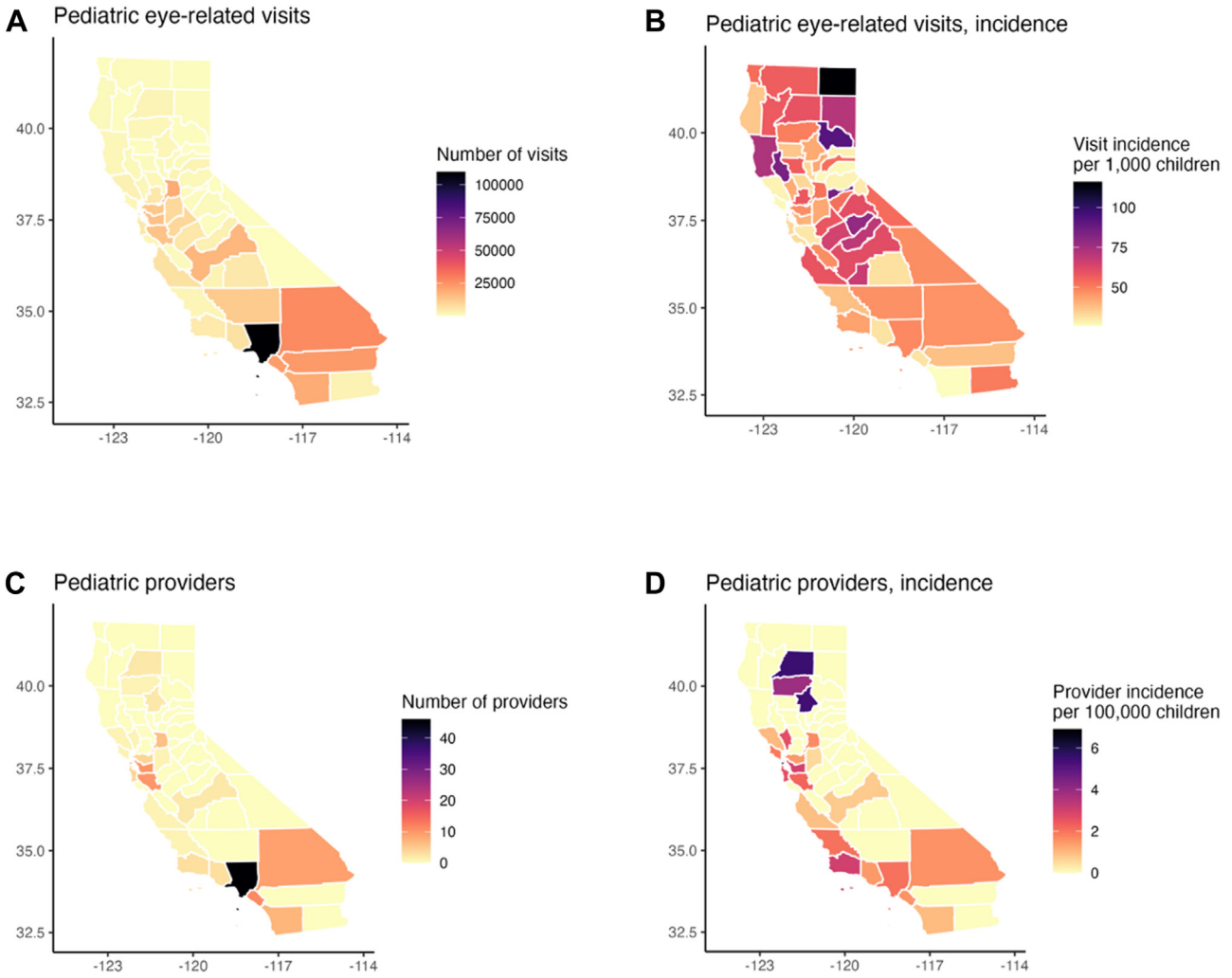


Figure 3. Geographic distribution of visits and providers by county. County-level **(A)** pediatric eye-related emergency department (ED) visits over the study period, **(B)** 10-year incidence of eye-related visits per 1000 children, **(C)** number of pediatric providers, and **(D)** number of pediatric providers per 100 000 children.

population) without any pediatric eye provider. Inland regions of the state were most affected by low provider density, consistent with prior studies describing a greater increase in ED utilization over time in rural compared with urban communities.[12]

In addition to pediatric eye provider density, we also identified several social drivers of health (SDoH) associated with eye-related ED visits. Social drivers of health refers to nonmedical factors that influence health outcomes, including income, education, employment status, social inclusion, and access to affordable health services.[13] In our study, population-level factors associated with more eye-related ED visits included unemployment, percentage of households on food stamps, and percentage of families

Table 4. Most Common Diagnoses (Prevalence and per 10 000 Children)

Primary Diagnosis	Number of Visits	Proportion of All Study Visits	Mean Yearly Number of Visits/10 000 Children
Conjunctivitis	250 028	63.8	27.6
Chalazion/Blepharitis	56 389	14.4	6.2
Other disorders of the eye and adnexa	13 070	3.3	1.4
Ocular pain	10 081	2.6	1.1
Subconjunctival hemorrhage	6153	1.6	0.7
Other visual disturbances	4730	1.2	0.5
Orbital cellulitis	4621	1.2	0.5

Table 5. Poisson Regression Evaluating Zip Code Level Sociodemographic Factors Associated with Eye-Related ED Visits and Pediatric Provider Density

Variable	Eye-Related ED Visits			Provider Density		
	Rate Ratio	*95% CI*	*P Value*	*Rate Ratio*	*95% CI*	*P Value*
High school graduate (per 10% increase)	1.15	1.14–1.17	<0.001	0.61	0.33–1.13	0.11
Bachelor's degree (per 10% increase)	0.91	0.90–0.92	<0.001	1.98	1.17–3.33	0.01
Foreign born (per 10% increase)	1.21	1.20–1.22	<0.001	1.29	0.74–2.25	0.38
Non-English language at home (per 10% increase)	0.89	0.88–0.89	<0.001	0.81	0.54–1.21	0.31
Unemployed (per 10% increase)	1.12	1.10–1.15	<0.001	0.52	0.09–2.92	0.46
Median household income (per $10 000 increase)	0.90	0.90–0.90	<0.001	0.80	0.72–0.88	<0.001
Households on food stamps or SNAP (per 10% increase)	1.02	1.01–1.03	<0.001	0.73	0.32–1.64	0.45
Families below federal poverty line (per 10% increase)	1.06	1.05–1.07	<0.001	1.08	0.53–2.07	0.83
Rural–urban commuting area code (per 1 point increase in RUCA score)	1.01	1.01–1.01	<0.001	0.71	0.39–0.99	0.13
Distance to nearest pediatric provider (per 10 km increase)	0.98	0.98–0.98	<0.001	-	-	-

CI = confidence interval; ED = emergency department; SNAP = supplemental nutrition assistance program.

below the poverty line. This is consistent with the existing literature showing that individuals with higher social risks may be more likely to visit the ED.[14,15] Insurance status has also been used as a surrogate for SDoH, with studies indicating the diverging impacts of the same treatments administered to patients with varying insurance statuses.[16] Children with public insurance represented 71.3% of eye-related ED visits despite the fact that only 44.7% of California children hold such insurance.[17] While race/ethnicity itself is not an SDoH, the racism faced by millions of non-White Americans is considered an SDoH as it is a key nonmedical factor that shapes health outcomes.[18] We found that Black and Hispanic Californians had disproportionately high levels of eye-related ED visits, as these populations constituted 12.7% and 55.9% of ED visits while only composing 7.1% and 39.4% of the state's population, respectively.[19] In contrast, patients who were Asian or White made up a smaller proportion of the ED visits relative to California's demographics.

While we identified several sociodemographic factors associated with eye-related ED visit incidence, fewer were associated with pediatric eye specialist density, where the only predictive factors were percentage of residents with a bachelor's degree and median household income. Interestingly, lower household income was associated with a greater number of pediatric eye specialists, though this may be

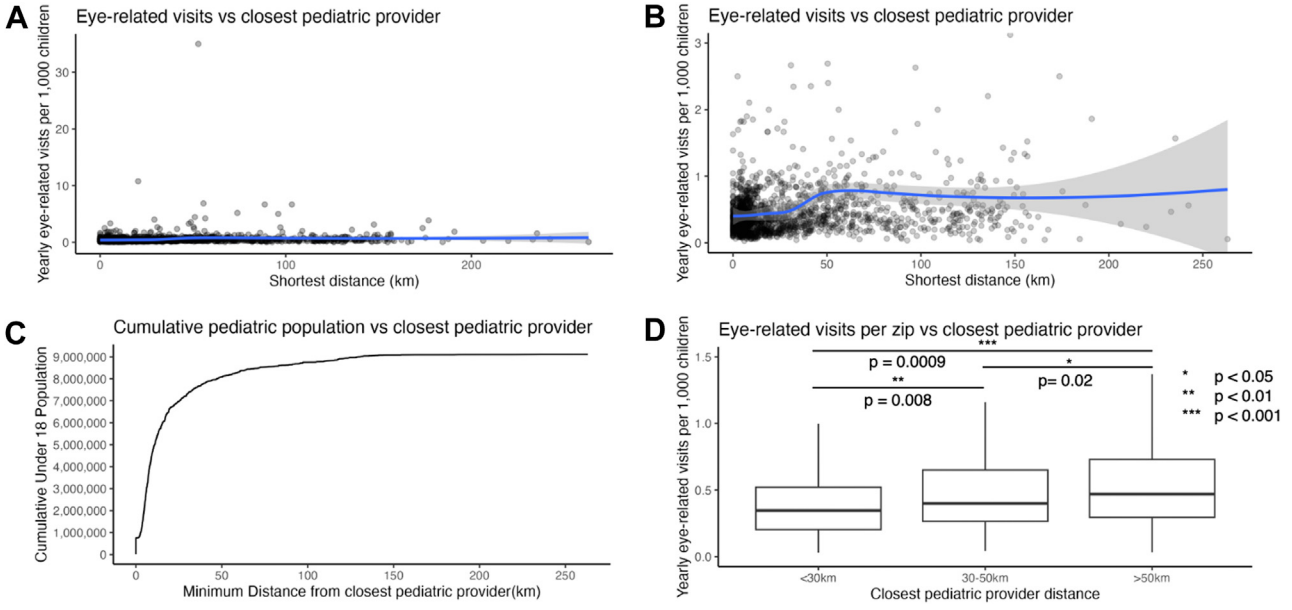


Figure 4. Eye-related visits and distance to provider. **A,** The yearly incidence of eye-related visits over the study period per zip code, plotted against the distance between the centroids of each zip code and the nearest pediatric provider's zip code (km). A local regression (locally estimated scatterplot smoothing) curve was fit to the data. **B,** Zoomed-in version of (A) given outliers in A. **C,** Cumulative pediatric population of each zip code graphed as a function of distance to the nearest pediatric provider. **D,** After grouping zip codes into ranges of pediatric provider distance (<30 km, between 30 and 50 km, and >50 km), a Welch 2-sample *t* test was used to calculate significant differences between each group.

explained by large hospitals and universities occupying smaller zip codes with relatively low median incomes. We used a multivariate Poisson regression model to identify these predictors, though this relies on aggregate sociodemographic data, which cannot account for within-zip code heterogeneity with regard to factors such as household income and parental education. Further research, including qualitative studies to assess reasons driving ED utilization and longitudinal studies to establish temporal relationships, is needed to deepen the understanding of these associations and inform targeted interventions.

This study underscores the need to address geographic disparities in access to pediatric eye specialists, which result from complex socioeconomic factors influencing eye providers' location. Disparities in access to care have tangible manifestations in patient behavior and ED utilization. Measures aimed at community engagement, increasing the accessibility of current pediatric eye specialists, and preparing a new generation of pediatric eye specialists must be considered in order to address these issues. While increasing the pediatric eye provider workforce is a large task, there are other actionable items that could potentially address aspects of this problem. Educational campaigns centered on pediatric eye health have been effective when they engage family units, and schools are often a common location to deploy such programming.[20] Empowering primary care providers to educate families on the presentation and treatment of common eye conditions may also mitigate familial concerns. Another approach could involve expanding the use of telemedicine by pediatric eye providers, and there is emerging evidence to support the use of telemedicine in pediatric ophthalmology.[21,22] Finally, exploring innovative care models such as mobile eye clinics or partnerships with other school-based health interventions could help improve access in underserved areas.

This study had several limitations. Given the nature of the data, we used zip codes to characterize geographic differences. Such aggregate data are subject to ecological fallacy and cannot be used to assess associations at an individual patient level. Moreover, zip codes vary widely in geographic area and population size, which can create outliers. We aimed to minimize this effect by excluding zip codes with a small number of children and also conducting certain analyses after aggregating at the county level. The pediatric eye provider data were gathered from the websites of public, professional organizations that allow providers to indicate if they would like their information publicized and, as such, may not be complete. We focused in this study specifically on pediatric eye care providers but acknowledge that other ophthalmologists and optometrists provide care for children as well. Finally, it is possible that other sociodemographic factors such as parental health literacy may drive increased ED utilization independent of eye provider availability.[23] Despite these limitations, the large and longitudinal nature of the dataset provides meaningful insight into the ways in which children receive eye-related ED care.

Lack of access to pediatric eye providers may be a driver of ED visits for nonemergent eye-related concerns. Expanding access to pediatric eye care providers could decrease ED utilization, and future efforts should focus on expanding and optimizing the pediatric eye care workforce.

Footnotes and Disclosures

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Analysis and interpretation: Oatts, Xu, Cavuoto, Keenan

Obtained funding: Oatts, Nesemann

Overall responsibility: Oatts, Xu, Morse, Nesemann, Cavuoto, Keenan

Abbreviations and Acronyms:
ACS = American Community Survey; **ED** = emergency department; **SDoH** = social drivers of health.

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